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Review text:

All hypergraphs considered have vertex set $[n] := \{1, \dots, n\}$ and are identified with their edge sets. A family $\mathcal{G}$ of r-uniform hypergraphs (r-graphs) on $[n]$ is said to be difference-isomorphic if $G_1 \setminus G_2$ is isomorphic to $G_2 \setminus G_1$ for all $G_1, G_2 \in \mathcal{G}$. For graphs, the problem of determining the maximum size of a difference-isomorphic family was raised by Alon–Gujgiczer–Körner–Milojević–Simonyi [SIAM J. Discrete Math. **37** (2023), no. 1, 379–403; MR4554523] in their work on graph codes, where examples of size $n^{(1+o(1))n}$ were given.

It turns out that much larger families exist. Let ψ be an involution of $[n]$ with at most one fixed point, and let $f_r(n)$ be the number of 2-cycles of the permutation of $\binom{[n]}{r}$ induced by ψ; explicitly, $f_r(n) = \frac{1}{2}\binom{n}{r}$ if r is odd and n is even, and $f_r(n) = \frac{1}{2}\left(\binom{n}{r} - \binom{\lfloor n/2 \rfloor}{\lfloor r/2 \rfloor}\right)$ otherwise. The family of all r-graphs that contain exactly one edge from each 2-cycle $\{e, \psi(e)\}$ and no other edges is difference-isomorphic, since ψ maps $G_1 \setminus G_2$ onto $G_2 \setminus G_1$ for any two members G_1, G_2, and it has size $2^{f_r(n)}$. The authors show that this is optimal: for each $r \geq 2$ and n sufficiently large, every difference-isomorphic family of r-graphs on $[n]$ has size at most $2^{f_r(n)}$. In particular, the answer to the question of Alon–Gujgiczer–Körner–Milojević–Simonyi (loc. cit.) is $2^{\frac{1}{2}\left(\binom{n}{2} - \lfloor n/2 \rfloor\right)}$ for all sufficiently large n.

Call $\mathcal{G}$ an involution clique if there is a single involution ψ of $[n]$ that maps $G_1 \setminus G_2$ onto $G_2 \setminus G_1$ for all $G_1, G_2 \in \mathcal{G}$. The authors also prove a stability result: if a difference-isomorphic family $\mathcal{G}$ is not an involution clique, then $|\mathcal{G}| \leq (1 - n^{-O(\sqrt{n})})\, 2^{f_2(n)}$ when $r = 2$, and $|\mathcal{G}| \leq e^{-\Omega(n^{r-2})}\, 2^{f_r(n)}$ when $r \geq 3$. The

bound for $r \geq 3$ is tight up to the constant in the exponent, and it is left open whether there is a difference-isomorphic family of graphs of size $(1-o(1))\,2^{f_2(n)}$ that is not an involution clique. The proof rests on bounding, for a fixed r-graph G and permutations φ, ψ of $[n]$, the number of pairs (G_1, G_2) of r-graphs related to G through φ and to each other through ψ; this number is shown to be negligible unless ψ is an involution that nearly coincides with φ. These estimates, together with further structural and counting arguments, yield the stability theorem.

The bound $2^{f_r(n)}$ does not hold for $n = r+1$, and the authors ask for which values of n, in terms of r, the maximum size of a difference-isomorphic family of r-graphs on $[n]$ equals $2^{f_r(n)}$.

Some minor typos: on page 26, the bound of Ellis–Filmus–Friedgut should read $2^{\binom{n}{2}-3}$; on page 32, $\binom{[n]}{2}$ and $2^{f_2(n)}$ should be $\binom{[n]}{r}$ and $2^{f_r(n)}$, respectively; in the last line of the proof of Lemma 3.10, "Item (3)" should be "Item (2)"; on page 45, $n^{-\binom{n}{r-2}/250}$ should be $e^{-\binom{n}{r-2}/250}$; in Claim 3.22, $G \in \mathcal{G} \setminus \mathcal{G}_0$ should be $G \in \mathcal{G}_0$, and $u_i y$ should be $v_i y$ in its proof; in Case 2.1 of the proof of Lemma 3.24, $\mathcal{F}$ should be $\bigcup_{\varphi \in P_3} (N_\varphi(G_0) \cap \mathcal{G}')$; and in the last condition defining G_0 in the appendix, $e \notin G$ should be $e \notin G_0$.